

Neurochemistry

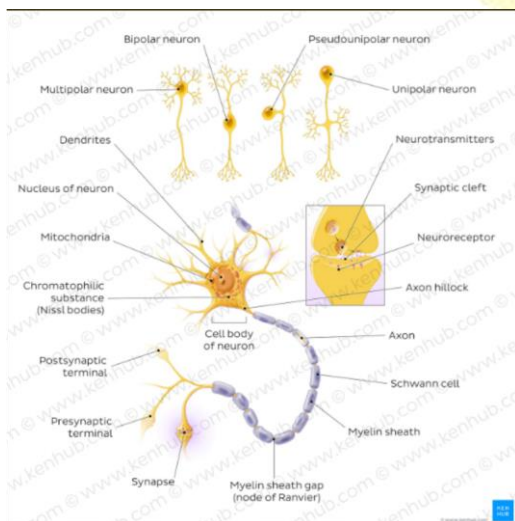
Neurotransmitters:-

Neurotransmitters are chemical messengers that your body can't function without. Their job is to carry chemical signals ("messages") from one neuron (nerve cell) to the next target cell. The next target cell can be another nerve cell, a muscle cell or a gland.

Neurotransmitters are chemical messengers that carry, boost, and balance signals between neurons (also known as nerve cells) and target cells throughout the body. These target cells may be in glands, muscles, or other neurons.

Billions of neurotransmitter molecules constantly work to keep our brain functioning, managing everything from breathing to heartbeat to learning and concentration levels. They can also affect various psychological functions such as fear, mood, pleasure, and joy.

Some common neurotransmitters in the brain and body include serotonin, dopamine, glutamate, epinephrine, norepinephrine, and endorphins.



How Neurotransmitters Work?

In order for neurons to send messages throughout the body, they need to be able to communicate with one another to transmit signals. However, neurons are not simply connected to one another.

At the end of each neuron is a tiny gap called a synapse, and to communicate with the next cell, the signal needs to be able to cross this small space. This occurs through a process known as neurotransmission.

In most cases, a neurotransmitter is released from what's known as the axon terminal after an action potential has reached the synapse, a place where neurons can transmit signals to each other.

When an electrical signal reaches the end of a neuron, it triggers the release of small sacs called vesicles containing neurotransmitters. These sacs spill their contents into the synapse, where the neurotransmitters then move across the gap toward the neighboring cells. These cells contain receptors where the neurotransmitters can bind and trigger changes in the cells.

After release, the neurotransmitter crosses the synaptic gap and attaches to the receptor site on the other neuron, either exciting or inhibiting the receiving neuron, depending on the neurotransmitter.

Receptors and transmitters act like a lock- and - key system. Just as it takes the right key to open a specific lock, a neurotransmitter (the key) will only bind to a specific receptor (the lock). If the neurotransmitter is able to work on the receptor site, it triggers changes in the receiving cell.

Inactivation of Neurotransmitters:-

Once the neurotransmitter has had the designed effect, its activity can be stopped by three mechanism:

Degradation: An enzyme changes the structure of the neurotransmitter so it can't be recognized by the receptor.

Diffusion: The neurotransmitter drifts away from the receptor

Reuptake: The whole neurotransmitter molecule is taken back up by the axon of the neuron that released it.

Criteria For Neurotransmitters:

The actual identification of neurotransmitters can actually be quite difficult. While scientists can observe the vehicles containing neurotransmitters, figuring out what chemicals are stored in the vesicles is not quite so simple.

Because of this, neuroscientists have developed a number of guidelines for determining whether or not a chemical should be defined as a neurotransmitter.

Presence of the chemical within the cell. The chemical is either synthesized in the neuron or otherwise found in it.

Stimulus- dependent release. It is released in appropriate quantities by the neuron upon stimulation.

Action on the postsynaptic cell. The chemical must be released by the presynaptic neuron, and the postsynaptic neuron must contain receptors that the chemical will bind to.

A mechanism for removal. A specific mechanism exists to remove the chemical from its site of activation after its work is done.

Classification of Neurotransmitters:-

Excitatory Neurotransmitters: These types of neurotransmitters have excitatory effects on the neuron, meaning they increase the likelihood that the

neuron will fire an action potential. Some of the major excitatory neurotransmitters include epinephrine and norepinephrine.

Inhibitory Neurotransmitters: These types of neurotransmitters have inhibitory effects on the neuron; they decrease the likelihood that the neuron will fire an action potential. Some major inhibitory neurotransmitters include serotonin and gamma- aminobutyric acid (GABA).

Modulatory Neurotransmitter: These neurotransmitters, often referred to as neuromodulators, are capable of affecting a larger number of neurons at the same time. These neuromodulators also influence the effects of other chemical messengers.

Types of Neurotransmitters:

There are a number of different ways to classify and categorize neurotransmitters. In some instances, they are simply divided into monoamines, amino acids, and peptides.

Neurotransmitters can also categorized into one of six types;

Amino Acids:- Gamma- aminobutyric acid (GABA); This naturally occurring amino acid acts as the body's main inhibitory chemical messengers. GABA contributes to vision and motor control, also playing a role in the regulation of anxiety, function by increasing the efficiency of GABA neurotransmitters, which can increase feelings of relaxation and calm.

Glutamate:- The most plentiful neurotransmitter found in the nervous system, glutamate plays a role in cognitive functions such as memory and learning. Excessive amounts of glutamate can cause excitotoxicity caused by glutamate build-up is associated with some diseases and brain injuries including Alzheimer's disease, stroke, and epileptic seizures.

Peptides:-

Oxytocin: This powerful hormone acts as a neurotransmitter in the brain. It is produced by the hypothalamus and plays a role in social recognition, bonding, and sexual reproduction.

Endorphins: These neurotransmitters inhibit the transmission of pain signals and promote feelings of euphoria. These chemical messengers are produced naturally by the body in response to pain, but they can also be triggered by other activities such as aerobic exercise.

Monoamines:-

Epinephrine: Also known as adrenaline, epinephrine is considered both a hormone and a neurotransmitter. Generally, epinephrine is a stress hormone that is released by the adrenal system.

Norepinephrine: This naturally occurring chemical is a neurotransmitter that plays an important role in alertness and is involved in the body's fight-or-flight response. Its role is to help mobilize the body and brain to take action in times of danger or stress.

Histamine: This organic compound acts as a neurotransmitter in the brain and spinal cord. It plays a role in allergic reactions and is produced as part of the immune system's response to pathogens.

Dopamine: Commonly known as the feel-good neurotransmitter, dopamine is involved in reward, motivation, and addictions. Several types of addictive drugs increase dopamine levels in the brain. Parkinson's disease which is a degenerative disease that results in tremors and motor movement impairments, is caused by the loss of dopamine-generating neurons in the brain.

Serotonin: A hormone and neurotransmitter, serotonin plays an important role in regulating and modulating mood, sleep, anxiety, sexuality, and appetite.

Selective serotonin reuptake inhibitors (SSRIs) are a type of antidepressant medication commonly prescribed to treat depression, anxiety, panic disorder, and panic attacks.

Purines:-

Adenosine: This naturally occurring chemical acts as a neuromodulator in the brain and is involved in suppressing arousal and improving sleep.

Adenosine triphosphate (ATP): Considered to be the energy currency of life, ATP acts as a neurotransmitter in the central and peripheral nervous system. It plays a role in autonomic control, sensory transduction, and communication with glial cells. Some neurological problems including pain, trauma, and neurodegenerative disorders.

Gasotransmitters:-

Nitric oxide: This compound plays a role in affecting smooth muscles, relaxing them to allow blood vessels to dilate and increase blood flow to certain areas of the body.

Carbon monoxide: This colorless, odorless gas can have toxic and potentially fatal effects when people are exposed to high levels of the substance. However, it is also produced naturally by the body where it acts as a neurotransmitter that helps modulate the body's inflammatory response.

Acetylcholine:-

Acetylcholine: This is the only neurotransmitter in its class. Found in both the central and peripheral nervous system, it is the primary neurotransmitter associated with motor neurons. It plays a role in muscle movements as well as memory and learning.